

Ahsanullah University of Science and Technology

Department of Computer Science and Engineering

Assignment 1

Course No : CSE 4142
Course Title : Data Warehousing and Mining Lab

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Submitted to ,

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Section : A

Group : 1

Year/Semester : 4/1

Introduction

This assignment focuses on understanding the importance of data preprocessing, class balancing, and model building in machine learning. The objective is to create a custom dataset with specified attributes and class distribution, apply data transformations, address class imbalance, and use a classification model (Naive Bayes) to predict outcomes on test data. This practical task provides insight into real-world data handling techniques essential in data mining.

Methodology

i. Dataset Creation

A Dataset was created with 5 Attributes: 2 Numeric, 2 Nominal & 1 Class (3 Class Values).

```
@relation customer_decision
```

```
@attribute Age numeric
```

```
@attribute Income numeric
```

```
@attribute Gender {Male,Female}
```

```
@attribute City {Dhaka, Chittagong, Sylhet}
```

```
@attribute Decision {Buy, NotBuy, Consider}
```

ii. Instance Creation

20 instances were created with the following class distribution:

- 10 instances for class: Buy
- 6 instances for class: NotBuy
- 4 instances for class: Consider

```
@data
```

```
25, 26000.25, Male, Dhaka, Buy
```

```
30, 50000.75, Female, Dhaka, Buy
```

```
27, 30000, Female, Sylhet, Buy
```

```
35, 25000.50, Female, Chittagong, Buy
```

```
26, 15000, Male, Sylhet, Buy
```

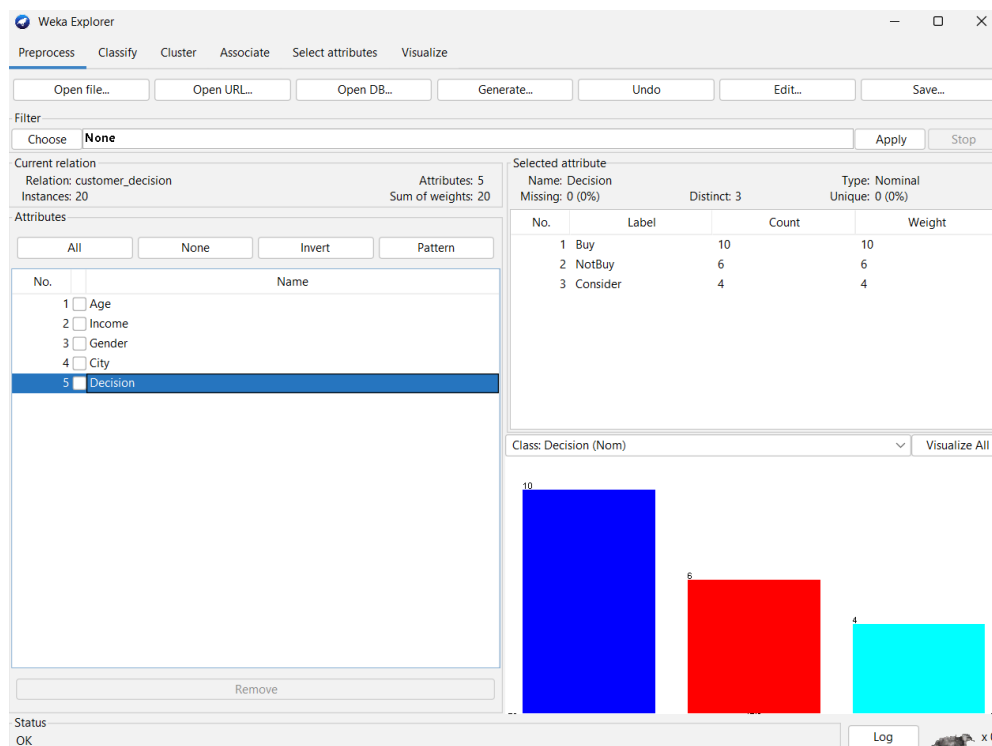
```
40, 60000, Male, Sylhet, Buy
```

```
55, 45000.75, Female, Chittagong, Buy
```

```
45, 20000.75, Male, Dhaka, Buy
```

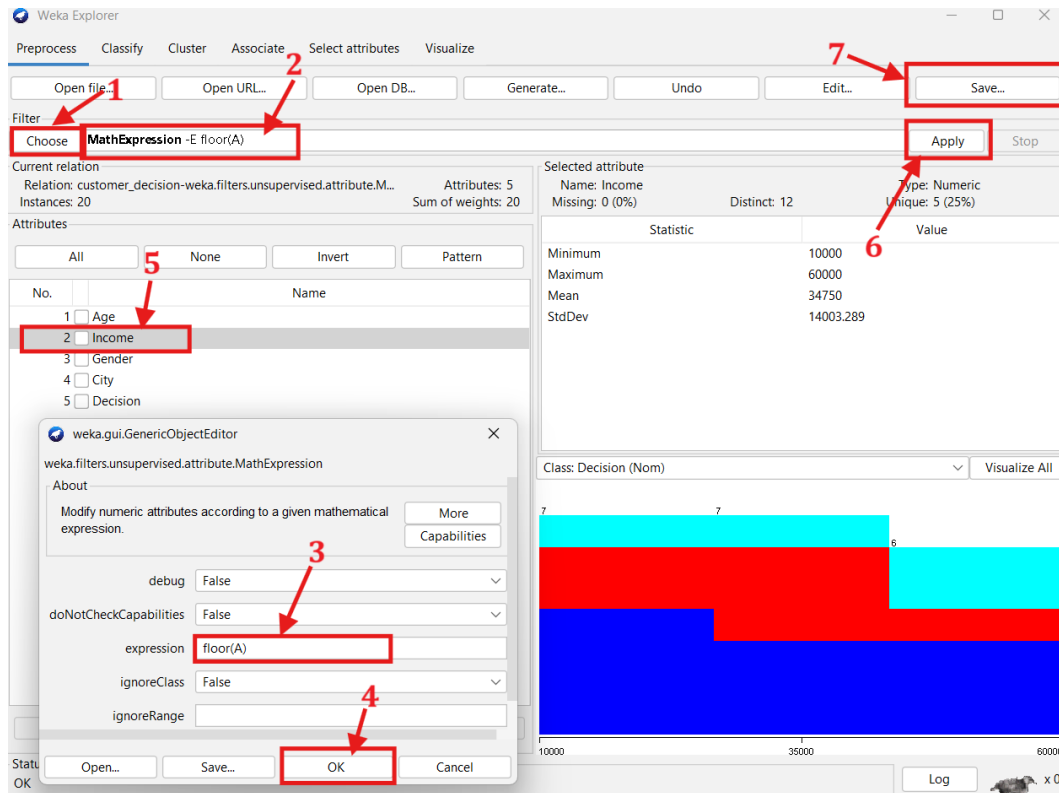
```
50, 36000, Female, Dhaka, Buy
```

32, 40000.00, Female, Chittagong, Buy
 30, 25000, Female, Sylhet, NotBuy
 20, 10000.50, Male, Dhaka, NotBuy
 45, 60000.25, Male, Chittagong, NotBuy
 35, 30000, Male, Chittagong, NotBuy
 50, 36000.00, Male, Dhaka, NotBuy
 27, 40000.75, Female, Chittagong, NotBuy
 25, 30000.50, Male, Chittagong, Consider
 40, 45000.75, Male, Dhaka, Consider
 65, 50000, Male, Dhaka, Consider
 50, 22000.25, Female, Sylhet, Consider



iii. Attribute Conversion

The **Income** attribute was originally a floating-point number. Using WEKA's MathExpression filter, the values were converted to integers using the floor function.



@relation customer_decision-
weka.filters.unsupervised.attribute.MathExpression-Efloor(A)

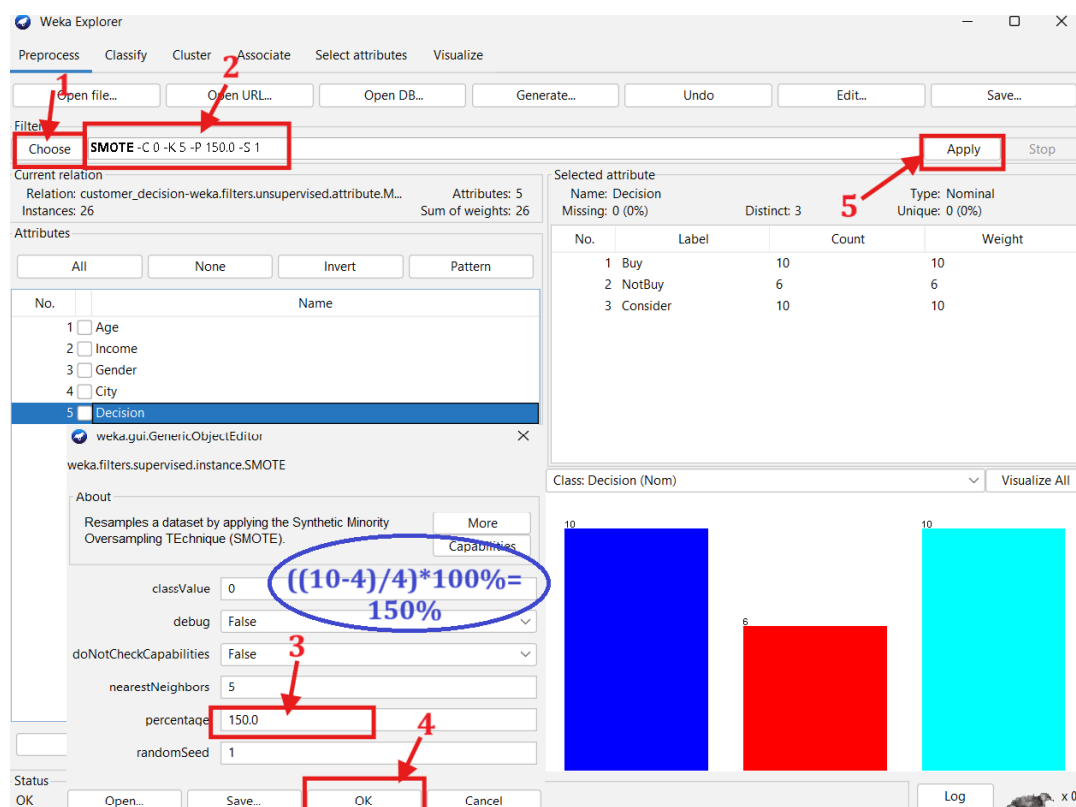
@attribute Age numeric
@attribute Income numeric
@attribute Gender {Male,Female}
@attribute City {Dhaka,Chittagong,Sylhet}
@attribute Decision {Buy,NotBuy,Consider}

@data
25,26000,Male,Dhaka,Buy
30,50000,Female,Dhaka,Buy
27,30000,Female,Sylhet,Buy
35,25000,Female,Chittagong,Buy
26,15000,Male,Sylhet,Buy
40,60000,Male,Sylhet,Buy
55,45000,Female,Chittagong,Buy
45,20000,Male,Dhaka,Buy
50,36000,Female,Dhaka,Buy
32,40000,Female,Chittagong,Buy

30,25000,Female,Sylhet,NotBuy
 20,10000,Male,Dhaka,NotBuy
 45,60000,Male,Chittagong,NotBuy
 35,30000,Male,Chittagong,NotBuy
 50,36000,Male,Dhaka,NotBuy
 27,40000,Female,Chittagong,NotBuy
 25,30000,Male,Chittagong,Consider
 40,45000,Male,Dhaka,Consider
 65,50000,Male,Dhaka,Consider
 50,22000,Female,Sylhet,Consider

iv. Class Balancing

To balance the dataset, SMOTE (Synthetic Minority Oversampling Technique) was applied.



Weka Explorer interface showing the SMOTE filter configuration for class balancing.

Selected attribute: Name: Decision, Missing: 0 (0%), Distinct: 3, Type: Nominal, Unique: 0 (0%).

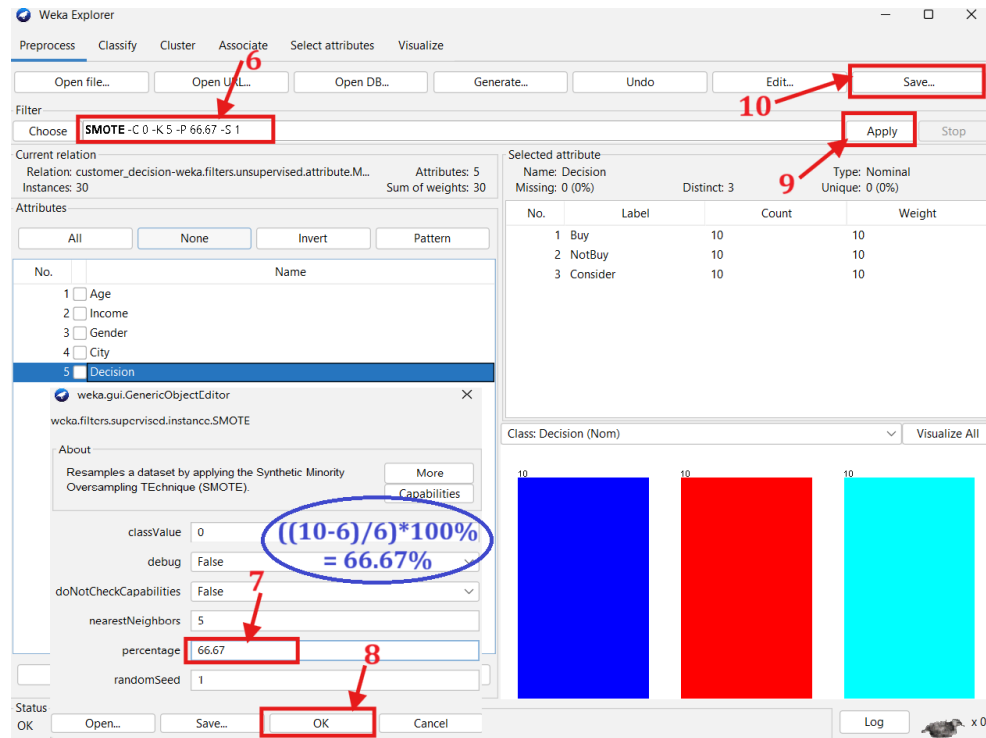
No.	Label	Count	Weight
1	Buy	10	10
2	NotBuy	6	6
3	Consider	10	10

Class: Decision (Nom) Visualize All

Handwritten formula: $((10-4)/4)*100\% = 150\%$

Configuration parameters:

- classValue: 0
- debug: False
- doNotCheckCapabilities: False
- nearestNeighbors: 5
- percentage: 150.0
- randomSeed: 1



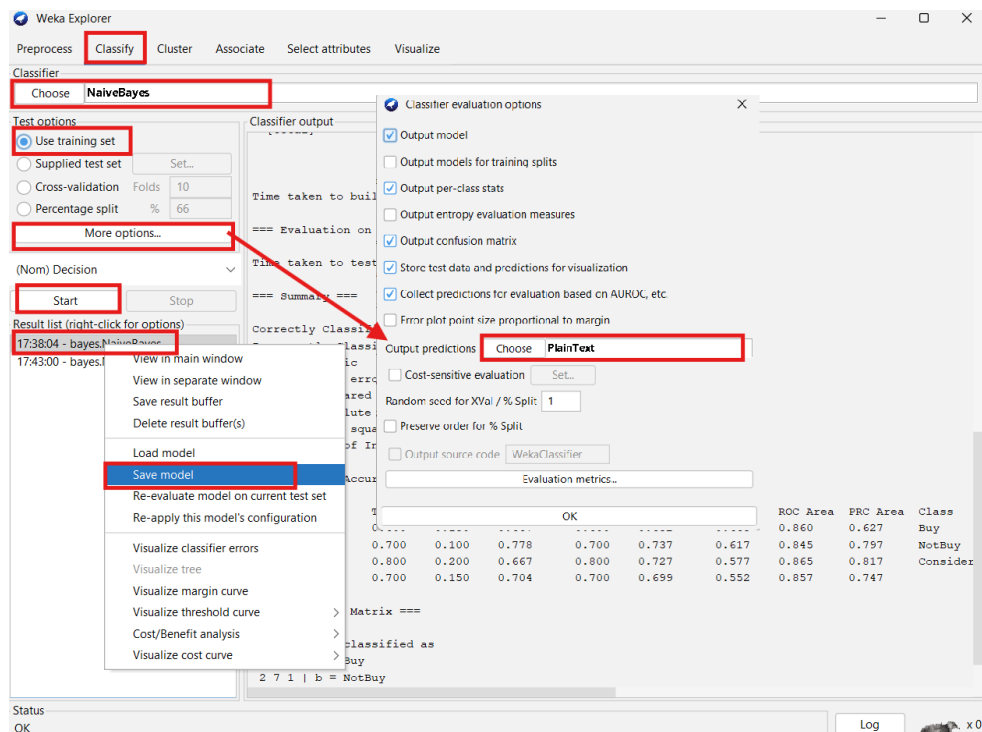
@relation customer_decision-
 weka.filters.unsupervised.attribute.MathExpression-Efloor(A)-
 weka.filters.supervised.instance.SMOTE-C0-K5-P150.0-S1-
 weka.filters.supervised.instance.SMOTE-C0-K5-P66.67-S1

@attribute Age numeric
 @attribute Income numeric
 @attribute Gender {Male,Female}
 @attribute City {Dhaka,Chittagong,Sylhet}
 @attribute Decision {Buy,NotBuy,Consider}

@data
 25,26000,Male,Dhaka,Buy
 30,50000,Female,Dhaka,Buy
 27,30000,Female,Sylhet,Buy
 35,25000,Female,Chittagong,Buy
 26,15000,Male,Sylhet,Buy
 40,60000,Male,Sylhet,Buy
 55,45000,Female,Chittagong,Buy
 45,20000,Male,Dhaka,Buy
 50,36000,Female,Dhaka,Buy
 32,40000,Female,Chittagong,Buy

30,25000,Female,Sylhet,NotBuy
 20,10000,Male,Dhaka,NotBuy
 45,60000,Male,Chittagong,NotBuy
 35,30000,Male,Chittagong,NotBuy
 50,36000,Male,Dhaka,NotBuy
 27,40000,Female,Chittagong,NotBuy
 25,30000,Male,Chittagong,Consider
 40,45000,Male,Dhaka,Consider
 65,50000,Male,Dhaka,Consider
 50,22000,Female,Sylhet,Consider
 30,27338,Male,Dhaka,Consider
 35,32291,Male,Dhaka,Consider
 63,49735,Male,Dhaka,Consider
 52,45430,Male,Dhaka,Consider
 54,36181,Male,Dhaka,Consider
 36,26666,Male,Dhaka,Consider
 34,58871,Male,Chittagong,NotBuy
 23,17978,Male,Chittagong,NotBuy
 21,13145,Male,Chittagong,NotBuy
 36,35043,Male,Chittagong,NotBuy

v. Naïve Bayes Model



Test Dataset:

@relation customer_decision_test

@attribute Age numeric

@attribute Income numeric

@attribute Gender {Male,Female}

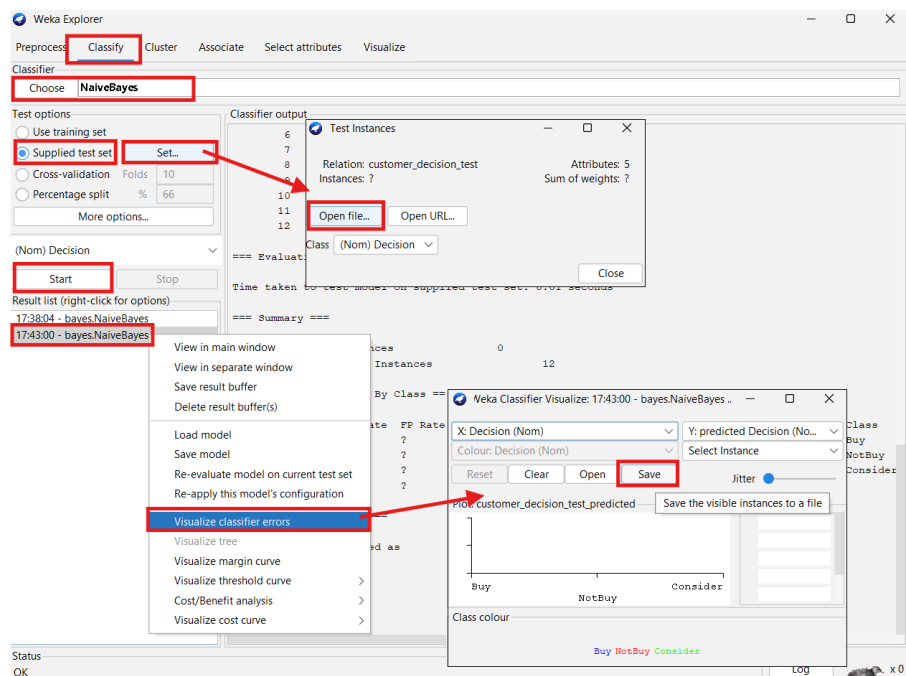
@attribute City {Dhaka, Chittagong, Sylhet}

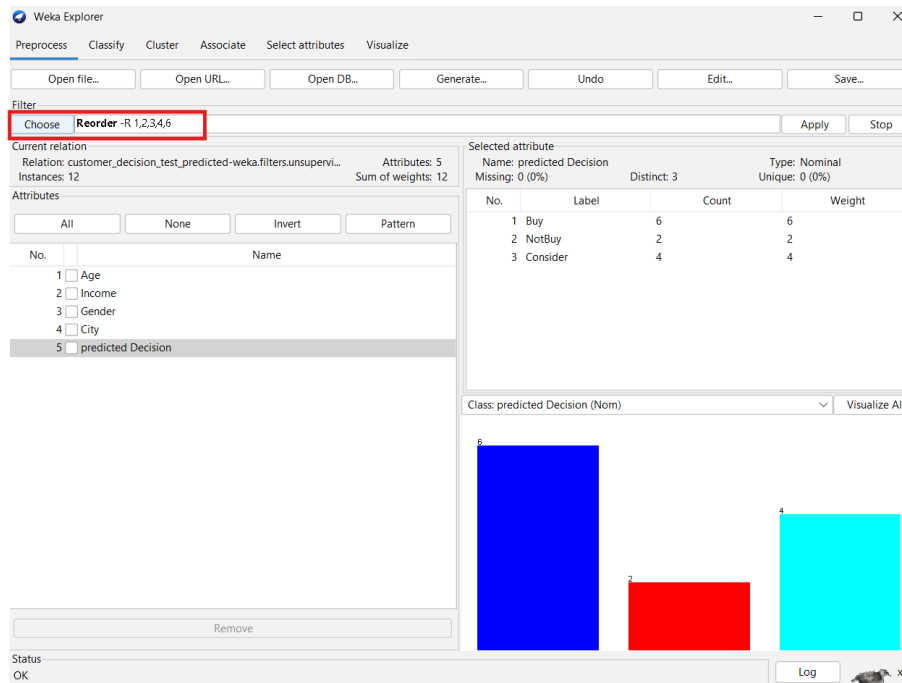
@attribute Decision {Buy, NotBuy, Consider}

@data

33, 28000, Female, Dhaka, ?
47, 42000, Male, Chittagong, ?
29, 33000, Female, Sylhet, ?
55, 38000, Male, Dhaka, ?
42, 46000, Female, Chittagong, ?
31, 21000, Female, Sylhet, ?
40, 35000, Female, Sylhet, ?
45, 28000, Male, Chittagong, ?
60, 32000, Female, Sylhet, ?
53, 37000, Female, Dhaka, ?
52, 52000, Female, Chittagong, ?
30, 28000, Male, Sylhet, ?

Modified Test Dataset with Predictions:





```
@relation 'customer_decision_test_predicted-
weka.filters.unsupervised.attribute.Reorder-R1,2,3,4,6'
```

```
@attribute Age numeric
```

```
@attribute Income numeric
```

```
@attribute Gender {Male,Female}
```

```
@attribute City {Dhaka,Chittagong,Sylhet}
```

```
@attribute 'predicted Decision' {Buy,NotBuy,Consider}
```

```
@data
```

```
33,28000,Female,Dhaka,Buy
47,42000,Male,Chittagong,Consider
29,33000,Female,Sylhet,Buy
55,38000,Male,Dhaka,Consider
42,46000,Female,Chittagong,Buy
31,21000,Female,Sylhet,Buy
40,35000,Female,Sylhet,Buy
45,28000,Male,Chittagong,NotBuy
60,32000,Female,Sylhet,Consider
53,37000,Female,Dhaka,Consider
52,52000,Female,Chittagong,Buy
30,28000,Male,Sylhet,NotBuy
```

Result Analysis

The preprocessing step of converting float income values to integers ensured consistency in data representation. Balancing the dataset using SMOTE corrected the initial class imbalance (Buy: 10, NotBuy: 6, Consider: 4), generating 10 instances for each class.

After training the Naive Bayes model on the balanced data, predictions were made on a test set of 12 instances. The model produced varied class predictions, indicating successful learning and generalization.

- SMOTE balanced the dataset for fair training.
- Income attribute conversion reduced noise.
- Naive Bayes effectively predicted diverse class labels.
- The process demonstrated an end-to-end data mining workflow using WEKA

Conclusion

This assignment provided hands-on experience in data preprocessing, dataset balancing, and classification model implementation. Through constructing a dataset, transforming attributes, applying SMOTE, and using the Naive Bayes algorithm, I have deepened my understanding of foundational data mining practices. These skills are crucial for building accurate and fair predictive models in real-world applications.